

EXPONENTS AND SURDS NOTES

a) Definitions and laws revision:

Definition/law	Example	Explanation
1. $x^0 = 1$	$(x^4 + 4)^0 \times 3^0 = 1 \times 1 = 1$	Any base raised to the power of zero is equal to 1. ($x \neq 0$ as 0^0 is undefined)
2. $x^{-a} = \frac{1}{x^a}$	$3x^{-2} = \frac{3}{x^2}$ and $\frac{3}{x^{-2}} = 3x^2$	A base raised to a negative exponent is equal to its reciprocal raised to the same positive exponent.
3. $x^a \times x^b = x^{a+b}$	$2^3 \times 2^2 \times 2 = 2^{3+2+1} = 2^6$	When multiplying like bases keep the bases the same and add the exponents.
4. $\frac{x^a}{x^b} = x^{a-b}$	$\frac{6x^6}{2x^2} = 3x^4$	When dividing like bases keep the base and subtract the exponent. Divide numbers as per normal.
5. $(x^a)^b = x^{ab}$	$(-2a^2b^3)^2 = (-2)^2 \times a^{2 \times 2} \times b^{3 \times 2} = 4a^4b^6$	When raising exponents to a power, keep the base and multiply the exponents.
6. $\left(\frac{x}{y}\right)^a = \frac{x^a}{y^a}$	$\left(\frac{a^3}{b}\right)^3 = \frac{a^9}{b^3}$	When a fraction is raised to an exponent, the numerator and denominator must be raised to that exponent.
7. $x^{\frac{m}{n}} = \sqrt[n]{x^m}$	$8^{\frac{2}{3}} = \sqrt[3]{8^2}$	If the exponent is a fraction, the base is raised to the numerator and rooted to the denominator.
8. $\sqrt[n]{xy} = \sqrt[n]{x} \times \sqrt[n]{y}$	$\sqrt{18} = \sqrt{9} \times \sqrt{2}$	When surds are multiplied they can be split apart and rooted individually.
9. $\sqrt[m]{\sqrt[n]{x}} = \sqrt[m \times n]{x}$	$\sqrt[6]{64} = \sqrt[3 \times 2]{64} = \sqrt[3]{\sqrt[2]{64}} = \sqrt[3]{8} = 2$	When taking the root of a root, it is the same as taking the single root to the product of both roots.

b) Addition and subtraction of fractions:

During addition and subtraction of fractions with variable exponents, Law 3 in reverse is used to split apart exponents. Then use factorisation to simplify the expression.

$$\begin{aligned}
 \text{a) } & \frac{3^{1+x} - 5 \cdot 3^x}{3^x \cdot 6} \\
 &= \frac{3^1 \cdot 3^x - 5 \cdot 3^x}{3^x \cdot 6} \quad \text{split bases} \\
 &= \frac{3^x(3-5)}{3^x \cdot 6} \quad \text{Find HCF and factorise} \\
 &= \frac{-2}{6} \\
 &= -\frac{1}{3} \quad \text{Simplify}
 \end{aligned}$$

c) Simplification of surds:

A surd is the root of a number that would result in an irrational number.

For example: $\sqrt{3}$ is a surd as the answer is irrational.

$\sqrt{9}$ would not be referred to as a surd as it has a rational answer (3).

$$\begin{aligned}
 \text{a) } & \sqrt{12} - \sqrt{48} + \sqrt{75} \\
 &= \sqrt{4 \times 3} - \sqrt{16 \times 3} + \sqrt{25 \times 3} \quad (\text{break apart into perfect square and another number}) \\
 &= 2\sqrt{3} - 4\sqrt{3} + 5\sqrt{3} \quad (\text{square root}) \\
 &= 3\sqrt{3} \quad (\text{simplify})
 \end{aligned}$$

$$\begin{aligned} \text{b) } & (\sqrt{5} - 2)(\sqrt{5} + 2) \quad (\text{DOTS}) \\ & = 5 - 4 \\ & = 1 \end{aligned}$$

d) Simple equations involving surds:

Steps:

- Isolate the surd.
- Square both sides.
- Solve for x .
- Check your answer.

e) Equations with rational exponents:

These are exponents with fractions.

If an equation is in the form: $x^{\frac{a}{b}} = y$

- There will be a positive and negative solution if a is even and b odd.
- There will be one solution if a is odd.

$$\begin{aligned} 1) \quad & x^{\frac{2}{3}} - 16 = 0 \\ & x^{\frac{2}{3}} = 2^4 \quad (a \text{ is even } b \text{ is odd } \therefore \pm \text{ solutions}) \\ & (x^{\frac{2}{3}})^{\frac{3}{2}} = \pm (2^4)^{\frac{3}{2}} \quad (\text{both sides raised to } \frac{3}{2}) \\ & x = \pm 2^6 \\ & x = \pm 64 \end{aligned}$$

$$\begin{aligned} 2) \quad & 3^{\frac{x}{4}} = 27 \\ & 3^{\frac{x}{4}} = 3^3 \quad (\text{convert 27 to a prime base}) \\ & \frac{x}{4} = 3 \quad (\text{if the bases are the same the exponents have to be equal}) \\ & x = 12 \end{aligned}$$

f) Equations involving factorising:

$$\begin{aligned} 1) \quad & 3^{x+2} + 3^{x+3} - 3^x = 105 \\ & 3^x \cdot 3^2 + 3^x \cdot 3^3 - 3^x = 105 \quad (\text{split bases apart}) \\ & 3^x(3^2 + 3^3 - 1) = 105 \quad (\text{factorise}) \\ & 3^x(35) = 105 \quad (\text{divide both sides by 35}) \\ & 3^x = 3 \\ & x = 1 \\ \\ 2) \quad & 3^{2x} - 10 \cdot 3^x + 9 = 0 \quad ((3^x)^2 \text{ use } k \text{ method}) \\ & \text{Let } k = 3^x \\ & k^2 - 10k + 9 = 0 \quad (\text{factorise}) \\ & (k - 9)(k - 1) = 0 \\ & 3^x = 9 \quad \text{or} \quad 3^x = 1 \quad (\text{get same bases on both sides of the equation}) \\ & 3^x = 3^2 \quad \text{or} \quad 3^x = 3^0 \quad (\text{any number raised to the power of zero equals 1}) \\ & x = 2 \quad \text{or} \quad x = 0 \end{aligned}$$